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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 06/11/2008 |
| **PI Name:** | Troian, Sandra M Ph.D. | | |
| **Project Title:** | MICROFLUIDS: ACTUATION BY MODULATION OF SURFACE STRESSES-FLOW MODELING TO DEVICE IMPLEMENTATION | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Interfacial phenomena | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 91125 | **Congressional District:** | 29 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2001 Physical Sciences 01-OBPR-08 |
| **Start Date:** | 01/28/2004 | **End Date:** | 01/26/2007 |
| **No. of Post Docs:** | 3 | **No. of PhD Degrees:** | 3 |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | 2 |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA GRC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE to 1/26/2007 per PI. Also, moved to California Institute of Technology from Princeton (10/06) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNC06GA44G ; NNC04GA52G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | The completed project has established two new developments: (a) an innovative microfluidic chip based on thermocapillary modulation of the speed and flow in liquid films and droplets for space flight applications and (b) optical sensing by an integrated thin film planar waveguide. This open format device is capable of a number of fluidic operations including control over droplet speed and direction, variable droplet volume and dispensing times for on chip liquid metering, multi droplet mixing, coalescence and scission. An integrated planar thin film waveguide beneath the transported droplets allows in situ sensing of droplet position and composition as well as direct measurement of reaction rate constants as a function of substrate temperature. In a side project related to thermocapillary actuation, it has also been shown how thermally generated surface stresses can be used to pattern polymeric nanofilms for use as photolithographic masks during the fabrication step on the microfluidic chip requiring the deposition of patterned self-asembled monolayers. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Commercial interest in microfluidic devices for propelling and manipulating liquids samples through hundreds of small channels has grown considerably over the past decade. The highly parallelized design for reaction, synthesis and analysis have reduced reagent volumes, waste products and cost. Device fabrication has benefited enormously from advances in silicon processing and MEMS technology as well as widespread use of inexpensive elastomeric polymers for surface patterning and encapsulation. As a result, microfluidic devices are being developed for such diverse applications as liquid transport in DNA microarrays, drug screening, optical display technologies, tunable fiber optic waveguides, thermal management of semiconducting devices and lasers, clinical and forensic analysis, and environmental monitoring for both earth and space based applications. We are particularly interested in developing hand held, low power fluidic platforms for both environmental cabin monitoring and and biodiagnostic assays necessary for evaluating astronaut health during space flight.<br>Most fluidic devices rely on internal flow within enclosed microchannels made of glass or silicon which is generated by application of electroosmotic, electrohydrodynamic, magnetohydrodynamic, centrifugal, capillary or pneumatic forces. Transport based on modulation of interfacial stresses, however, is especially effective in controlling the speed, shape and composition of liquid droplets and films for microfluidic devices since the surface to volume ratio is very large. Controlled modulation of interfacial forces has been demonstrated by other research groups using electrowetting, dielectrophoresis, and Marangoni stresses.<br><br>Our project has introduced a new microfluidic device for droplet and thin film actuation (including flow speed and direction, mixing, splitting) using thermocapillary stresses. These stresses are induced by activating linear arrays of embedded microheaters which can be tuned to present various surface thermal maps useful for various fluidic functions. This type of open architecture format allows direct accessibility to liquid samples for handling, sensing and diagnostic purposes. It is also a powerful method for fluidic actuation in the absence of significant gravitational forces. Lower operating voltages have also been demonstrated because the retentive frictional losses affecting microdroplets are only confined to the supporting liquid-solid boundary, which in conventional internal flows is present on all four sides. Gas-liquid interfaces are also effective as flexible capture surfaces for environmental applications targetted to airborne contaminants since these are readily attracted and adsorbed to the interface through electrostatic or van der Waals forces (via the liquid surface tension). The main disadvantage of open systems is evaporative mass loss which can be significantly reduced through encapsulation by a second liquid or by lowering the operating temperatures while maintaining the same thermal gradient since thermocapillary stresses are proportional to the local temperature gradient.<br><br>The significance of our most recent work in this area involves direct integration of evanescent wave sensing into our previous design. A planar waveguide was fabricated onto the microheater surface and the attenuation of light launched into this waveguide was used to monitor droplet position, dye concentration in bioassays upon reaction, and in-situ measurements of reaction rate constants as a function of temperature for key chromogenic bioreactions. This demonstration illustrates just a few of the capabilities possible by direct integration of optical sensing with “open platform” fluidic devices. |
| | The development of our novel microfluidic chip involved several four separate tasks as outlined briefly below. These tasks were necessitated by the following observations:<br>(i) The onset of movement and flow in an initially stationary droplet requires one overcome the retarding effects due to contact angle hysteresis. We therefore launched a task devoted to understand this phenomenon. We successful developed a model to help us estimate the pinning forces involved and what power to input to our device in order to dislodge and start moving a stationary droplet. This work is described in (1) below.<br><br>(ii) Our second task (2) describes the work involved in integrating a thin film planar waveguide into our microfluidic substrate in order to conduct in-situ optical sensing of droplets traversing the interrogration beam. This method of sensing relies on the diminution of the intensity of the evanescent wave whenever a droplet contains constituents which absorb light at the wavelength set by the totally internally reflected interrogation beam. This is described in (2) below.<br><br>(iii) The third task describes supporting theoretical work to try and understand how to maximize slip at liquid/solid boundaries in order to reduce the degree of friction inherent in small scale flows. We conducted both continuum and molecular level simulations to probe what surface patterns and non wetting coatings to use as friction reduction surfaces for our microfluidic device. This work is described in (3) below.<br><br>(iv) The fourth task describes additional supporting theoretical work to try and understand if the flow of thin films achieved with our microfluidic device is subject to any undesirable instabilities, like breakup or finger formation. In particular, we focus on the linear stability of thin film flows driven by thermocapillary stresses over topologically structured surfaces, as present in microfluidic devices. This work is described in (4) below.<br><br>(1) Effect of Contact Angle Hysteresis on Thermocapillary Droplet Actuation. Participants: Chen, Darhuber and Troian.<br><br>Open microfluidic devices based on actuation techniques such as electrowetting, dielectrophoresis, or thermocapillary stresses require controlled motion of small liquid droplets along the open surface of glass or silicon substrates. We explored the physical mechanisms affecting thermocapillary migration of droplets generated by surface temperature gradients on homogeneous silicon substrates coated by self assembling monolayers of OTS or DTS. These coated substrates presented a partially wetting surfaces to droplets of squalane (C30H62, 99%), n-decane (C10H22, 99%), n-dodecane (C12H26, 99%), n-tetradecane (C14H30, 99%), and n-hexadecane (C16H34, 99%). Using a combination of |

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| **Task Progress:** | experiment and a simplified two dimensional model incorporating the effects of capillary, thermocapillary, gravity and contact angle hysteretic forces, we investigated the behavior of the threshold force required for droplet mobilization and the speed after depinning as a function of the droplet size, the applied thermal gradient and the liquid material parameters like surface tension and viscosity. The experimental results were found to be well described by a generalized hydrodynamic model based on earlier work by Ford and Nadim [Phys. Fluids 6, 3183 (1994)]. The results of this study highlight the critical role of chemical or mechanical hysteresis and the need to reduce this retentive force for minimizing power requirements in microfluidic devices.<br><br>(2) Microfluidic Detection and Analysis by Integration of Thermocapillary Actuation with a Planar Thin Film Waveguide. Participants: Valentino and Troian.<br><br>We demonstrated a non-intrusive optical method for microfluidic detection and analysis based on evanescent wave sensing. The device consists of a planar thin film waveguide made of silicon nitride SiNx (refractive index n2=1.84) deposited by plasma enhanced chemical vapor deposition integrated with a microfluidic chip based on thermocapillary actuation flow. Microliter droplets were electronically transported and positioned over the waveguide surface by thermocapillary actuation. The attenuated intensity of propagating modes was used to detect droplet location, monitor dye concentration in aqueous solutions, and measure reaction rates with increasing surface temperature for a chromogenic biochemical assay. This project and the device integration achieved illustrate just a few of the capabilities made possible by direct integration of optical sensing with surface directed fluidic devices. As far as we know, this is the first time that an evanescent wave based sensor has been integrated into an open microfluidic device.<br><br>(3) Slip Behavior in Thin Films Subject to Planar Shear: Comparison Between Molecular Level Models and Continuum Based Simulations. Participants: Priezjev, Darhuber and Troian.<br><br>(i) We investigated the behavior of the slip length in Newtonian liquids subject to planar shear bounded by substrates with mixed boundary conditions. The upper wall, consisting of a homogenous surface of finite or vanishing slip, was translated at a constant speed parallel to a lower stationary wall, whose surface was patterned with an array of stripes representing alternating regions of no shear and finite or no slip. Velocity fields and effective slip lengths were computed for flow configurations either parallel or perpendicular to the stripes. Excellent agreement between the hydrodynamic and MD results was obtained when the normalized width of the slip regions, a/sigma > O(10) where sigma is the fluid molecular diameter characterizing the Lennard-Jones interaction potential. In this regime, the effective slip length increases monotonically with a/sigma toward an asymptotic value. For a/sigma < O(10) and transverse flow configurations, the non-homogeneous interaction potential along the lower wall constitutes a roughened type surface whose molecular scale corrugations strongly reduce the effective slip length below the hydrodynamic prediction. The translational symmetry for longitudinal flow eliminates the influence of molecular scale roughness; however, the reduced molecular ordering above the wetting regions of finite slip for small values of a/sigma increases the value of the effective slip length far above the hydrodynamic predictions. The strong correlation between the effective slip length and the liquid structure factor representative of the first fluid layer near the patterned wall illustrates the influence of molecular ordering effects on slip in non-inertial flows.<br><br>(ii) The influence of surface roughness on the slip behavior of a Newtonian liquid in steady planar shear was investigated using three different approaches, namely Stokes flow calculations, molecular dynamics (MD) simulations and a statistical mechanical model for the friction coefficient between a corrugated wall and the first liquid layer. These approaches were used to probe the behavior of the slip length as a function of the slope parameter ka =2 pi a/lamda, where a and lambda represent the amplitude and wavelength characterizing the periodic corrugation of the bounding surface. The molecular and continuum approaches both confirm a monotonic decay in the slip length with increasing ka but the rate of decay as well as the magnitude of the slip length obtained from the Stokes flow solutions exceed the MD predictions as the wall feature sizes approach the liquid molecular dimensions. In the limit of molecular-scale wall corrugation, a Green-Kubo analysis based on the fluctuation-dissipation theorem accurately reproduces the MD results for the behavior of the slip length as a function of the roughness amplitude a. In combination, these three approaches provide a detailed picture of the influence of periodic roughness on the slip length which spans multiple length scales ranging from molecular to macroscopic dimensions.<br><br>(4) Generalized linear stability of non-inertial coating flows over substrates with topographical features. Participants: Jeffrey M. Davis and Sandra M. Troian.<br><br>Actuation of thin liquid films along the surface of an open microfluidic device requires an understanding of the stability of non-inertial flows over surfaces with topological features. These topological features can represent bumps or indentations which are intentionally patterned onto the substrate for various purposes or which can occur accidentally as a result of various etching and deposition processes. We therefore investigated the transient evolution of perturbations to steady lubrication flow over a topographically patterned surface via a non-modal linear stability analysis of the non-normal disturbance operator. In contrast to the capillary ridges that form near moving contact lines in the forced spreading of thin liquid films on uniformly flat substrates, the stationary capillary ridges near trenches or elevations have only stable eigenvalues. Minimal transient amplification of perturbations occurs, regardless of the magnitude or steepness of the topographical features. The absence of transient amplification and stability of the ridge can be explained on physical grounds. By comparison to unstable ridge formation on smooth, flat, and homogeneous surfaces, the lack of closed, recirculating streamlines beneath the capillary ridge is directly linked to the linear (fingering) stability. |

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| **Bibliography Type:** | Description: (Last Updated: 03/26/2021) |
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| **Articles in Peer-reviewed Journals** | Davis JM, Kataoka DE, Troian SM. "Transient dynamics and structure of optimal excitations in thermocapillary spreading: Precursor film model." Physics of Fluids 2006 Sep;18(9):092101. http://dx.doi.org/10.1063/1.2345372 , Sep-2006 |
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| **Significant Media Coverage** | Editors of Analytical Chemistry journal. "Article on Troian method for integration of evanescent wave sensing with open microfluidic chip." Editorial Picks, Article on Troian method for integration of evanescent wave sensing with open microfluidic chip. Analytical Chemistry: Analytical Currents, 'Non-intrusive Droplet Analysis,' p. 241A, July 2006., Jul-2006 |